



**DCL-02**

**PAM TIME DIVISION  
MULTIPLEXING / DEMULTIPLEXING**

**EXPERIMENTAL MANUAL**



**.... A MESSAGE FROM**

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## **SAFETY RULES**

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that its integrity is checked with respect to its checklist. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In DCL-02 kit, signals are to be monitored with an oscilloscope as explained in the manual. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180Ω resistance in series with normal oscilloscope probe.

These kits must be employed only for the use for which it has been conceived, i.e.as educational equipment, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or malfunctioning in the trainer kit, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under following conditions,

- Ambient temperature: from 0 to 45 ° C.
- Relative humidity: from 20 to 80 %.

Avoid any immediate/significant change of temperature and humidity.

## **WARRANTY**

These kits are warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

**This warranty is subject to the following conditions and limitations.** The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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## **TECHNICAL SPECIFICATIONS**

**Input Channels:** 4 Numbers.

**Switching:** Time Division Multiplexing

**Modulation:** Pulse Amplitude Modulation.

**Onboard Sine Waves:** 2 KHz, 1 KHz, 500 Hz And 250 Hz

**Low Pass Filters:** 4th Order Butterworth Filters.

**Switch Faults:** 8 Switch Faults are Provided On Board to Study Different Effects on Circuit

**Interconnections:** 2mm Sockets and Connecting Chords Are Provided For Connection On Board

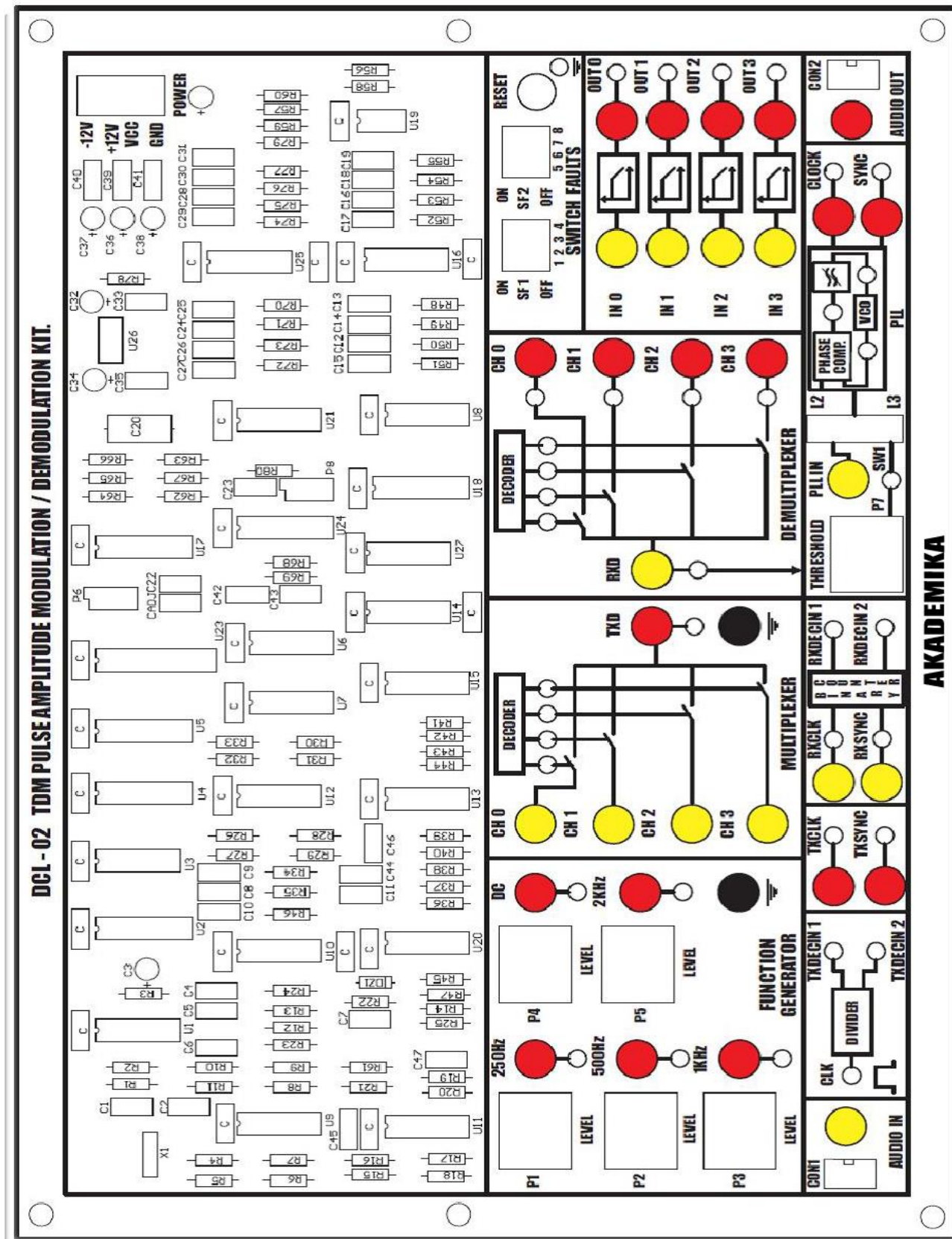
**Test points:** All Relevant Test Points Are Brought for Various Signals

**Power Supply:** -12v, +12v, +5v, GND

**Accessories:**

| <b>DCL -02</b> |                             |              | <b>Quantity</b> |
|----------------|-----------------------------|--------------|-----------------|
| 1              | Mini Links PCS2 Patch cord  | Male To Male | 10              |
| 2              | Short Links PCS2 Patch cord | Male To Male | 4               |
| 3              | DCL-02 EXPT Manual          |              | 1               |
| 4              | Power Supply                |              | 1               |
| 5              | Power cord                  |              | 1               |

## FUNCTIONAL BLOCK





## **INTRODUCTION**

**“DCL-02: TDM PULSE AMPLITUDE MODULATION / DEMODULATION KIT”** is a single board unit and it provides facilities for experimentation in the area of Time Division Multiplexing and Pulse Amplitude Modulation.

An onboard Function Generator provides four synchronized amplitude variable sine waves: 250Hz, 500Hz, 1 KHz, 2 KHz, and an amplitude variable DC level for generation of synchronization pulses. The 32 KHz-sampling clock and 8 KHz channel identification signal are also generated onboard.

Phase Locked Loop (PLL) logic circuitry is used to generate the clock and Channel Identification Information at the Receiver.

On each Receiver channel, a 4th order Low Pass Butterworth Filter is provided with cut-off frequencies of 3.4 kHz, to obtain reconstructed signals from the de-multiplexed and demodulated signals. The module provides facility for experimentation in 3 levels:

### **Level I**

The transmitter clock and Channel Identification Information is directly linked to the receiver.

### **Level II**

Only the channel identification information is directly linked to the receiver and the receiver clock is generated by a phase locked loop circuitry.

### **Level III**

Here one channel is used to send the synchronization pulses. The receiver derives the clock and channel identification information from the incoming stream of time division multiplexed pulses. A threshold comparator and phase locked loop circuitry is used to generate the clock and channel identification information at the receiver.

## **Principles of Time Division Multiplexing**

Pulse Modulation may be used to transmit analog information such as continuous speech or data. Here continuous waveforms are sampled at regular intervals. Information regarding the signal is transmitted only at the sampling times, together with the synchronizing pulse that may be required. At the receiving end, the original waveforms may be reconstructed from the information regarding the samples, if these are taken frequently enough. Despite the fact the information about the signal is not supplied continuously as in AM and FM, the resulting output at the receiver can have negligible distortion.

## **Pulse Amplitude Modulation (PAM)**

Pulse Amplitude Modulation is the analog form of Pulse Modulation. In PAM, the signal is sampled at regular intervals and the amplitude of each sample is made proportional to the amplitude of the signal at that instant of sampling. The pulses are then sent by either wire or cable or else are used to modulate a carrier. As shown in figure A, the two types of pulse amplitude modulation are double polarity PAM, which is self-explanatory and single polarity PAM, in which a fixed DC level is added to the signal to ensure that the pulses are always positive.

### **Time Division Multiplexing (TDM)**

One of the greatest benefits to be derived from sampling is that of Time Division Multiplexing (TDM). By inter-leaving samples of several source waveforms in time, it is possible to transmit enough information to a receiver, via only one channel to recover all message waveforms. This process is called Time Division Multiplexing (TDM). The conceptual implementation of the time multiplexing of  $N$  similar messages  $f_n(t)$   $n=1,2, N$  is illustrated in figure B. Sampled signals (Pulse Trains) for message  $f_1(t)$  and  $f_2(t)$  are shown in figure. The Pulse Train of second message is delayed slightly from the Train of first message to prevent overlap. Other message is treated similarly. When such  $N$  total Pulse Trains are combined, i.e., multiplexed, the waveform as shown in figure is obtained. The time allocated to one sample of one message is called a Time Slot. The time interval over which all messages are sampled at least once is called a Frame.

The portion of the Time Slot not used by any of the Sample Pulse is called the Guard Time. In a practical system, some Time Slots may be allocated to other function like Signaling, Monitoring, and Synchronization etc.

### **Synchronization**

To maintain proper positions of Sample Pulses in the Multiplexer, it is necessary to synchronize the Sampling Process. Because the sampling operations are usually electronic, there is typically a Clock Pulse Train. That serves as a reference for all samples. At the Receiving Station, a similar Clock Synchronization can be derived from the received waveforms by observing the Pulse Sequence over many pulses and averaging the pulses (in a closed loop with the Clock derived on the Voltage Controlled Oscillator).

Clock Synchronization does not guarantee that the proper sequence of samples is synchronized. Proper alignment of the Time Slot Sequence requires Frame Synchronization. Hence one or more Time Slots per Frame may be used to send Synchronization Information. For example, by placing a Special Pulse with larger amplitude than the largest expected Message Amplitude in TIME SLOT-1, the start of a Frame can easily be identified using a suitable Threshold Circuit.

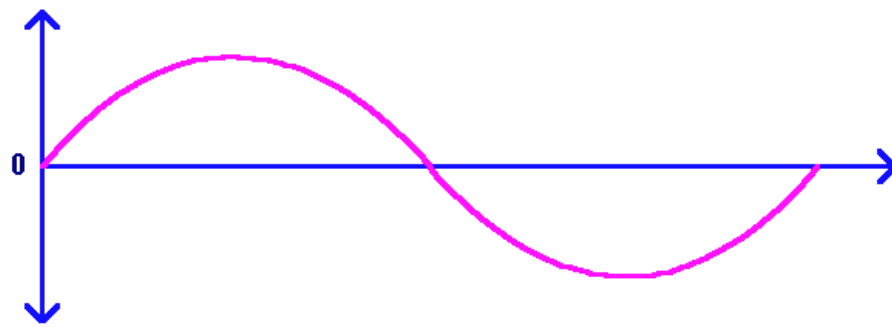
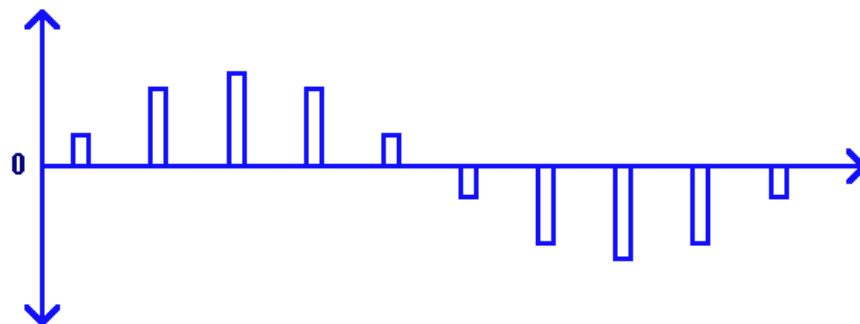
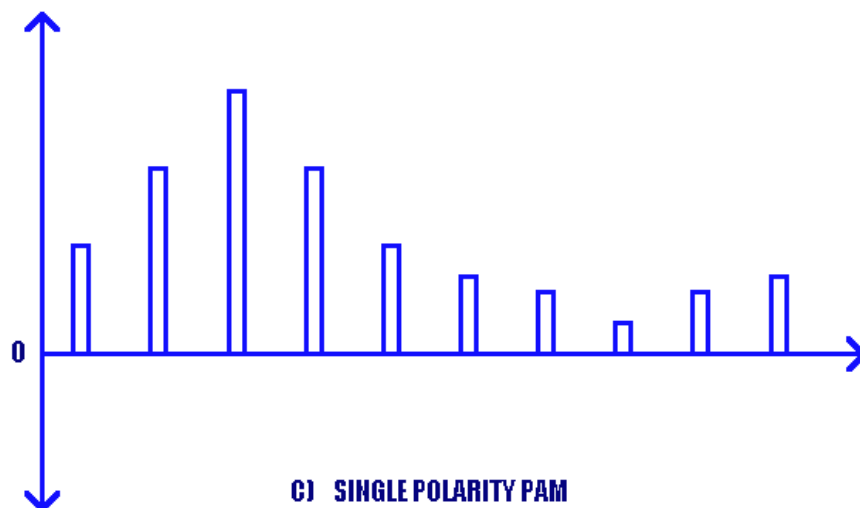
## **CIRCUIT DESCRIPTION**

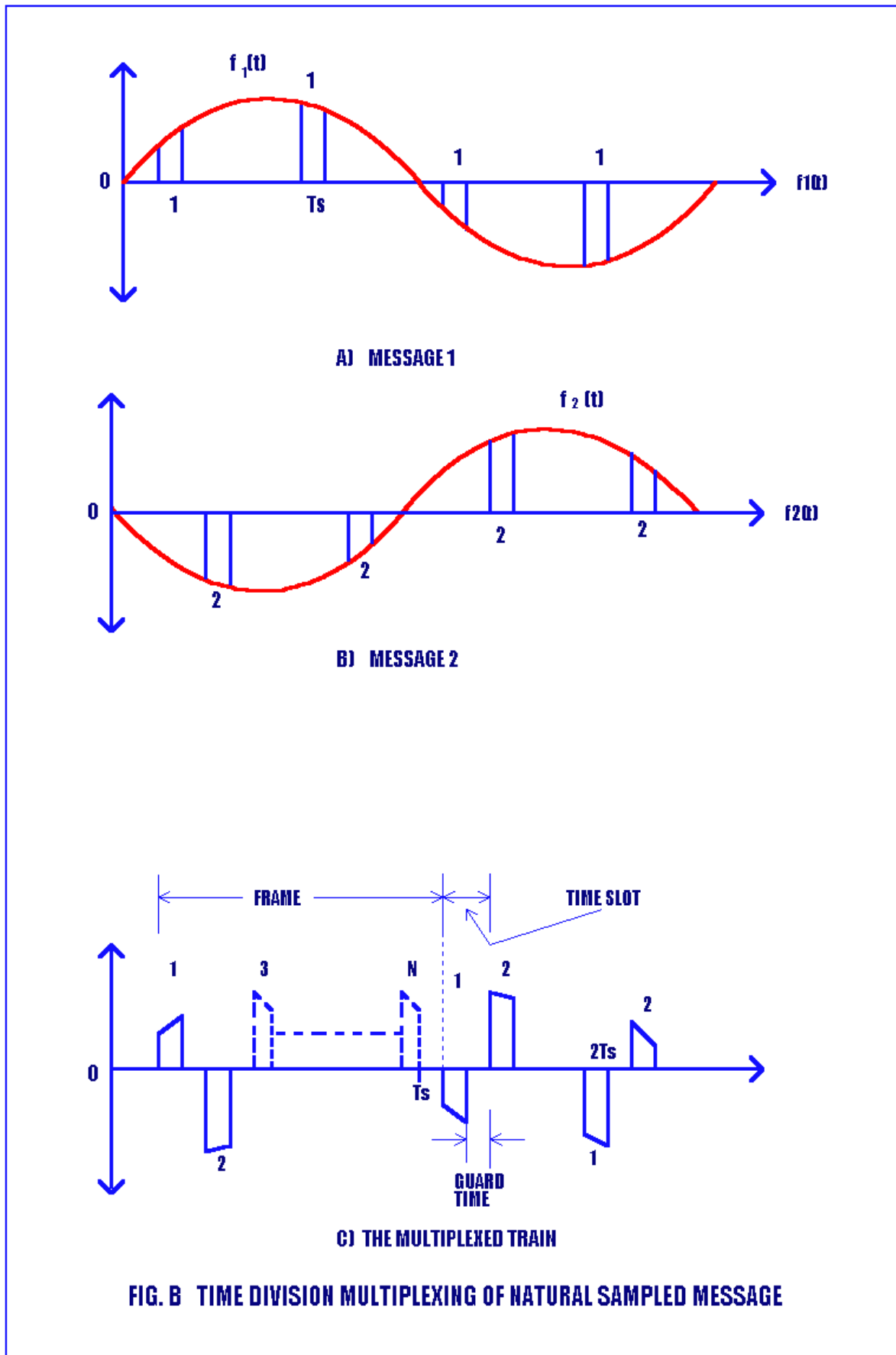
### **Clock Generation**

The 6.4 MHz crystal oscillators generate a 6.4 MHz clock. It is divided by 2 decade counters U2 (IC 74LS390) and 2 ripple counters U3 (IC 74LS393) to get the 32 KHz, 16 KHz, 8 KHz, and 4 KHz frequencies. 32 KHz frequency is used as a Sampling Clock and 8 KHz frequency is used for Channel Identification.

### **Onboard Function Generator**

This basically provides four Amplitude Variable (0-5 V-PP) synchronized sine waves, each 250Hz, 500Hz, 1KHz and 2KHz. And an amplitude variable DC level (0-5V).

**A) SIGNAL****B) DOUBLE POLARITY PAM****C) SINGLE POLARITY PAM****FIG.A WAVEFORMS FOR PULSE AMPLITUDE MODULATION**



The 6.4 MHz crystal oscillators generate a 6.4 MHz clock. It is divided by 2 decade counters U2 (IC 74LS390) and 2 ripple counters U3 (IC 74LS393) to get the 32KHz, 16KHz, 8KHz, 4KHz frequencies. This signal is fed to serial to parallel shift register U9, U11, U12, U13 (IC 74HC164), which generates the sine wave, by serial shift operations. The serial to parallel shift registers (IC 74HC164) have resistive ladder network at the output. For 16 shifts of the register, one stare case sine wave is produced. So if a 16 KHz clock is fed to the shift register, 1 KHz, sine wave is generated. Thus by using the clock 32 KHz, 16 KHz, 8 KHz, 4 KHz, sine waves of 2 KHz, 1 KHz, 500 Hz and 250 Hz are generated respectively. The active filter U10 (IC LM324) at the output suppresses the ripple and also takes care of the impedance matching. This gives smooth sine waves. DC level is regulated by the 5V zener.

### Transmitter

The Transmitter Section consists of four Analog Input Channels with 4-pole Integrated Circuit Analog Switch U15 (IC 4066) that provides Sampling and Time Division Multiplexing of each channel using Pulse Amplitude Modulation techniques.

The 32 KHz sampling clock is fed to the cascade of two counters. The two counters are configured in such a way that the outputs vary from 00 to 11. As the control input of the decoder varies from 00 to 11, the signal C1, C2, C3, C4 selects the channel of the analog switch U15 (IC 4066) and the information of that channel is transmitted. Thus TDM is achieved effectively. The unity gain amplifiers are provided to give proper impedance matching

### Receiver

The Receiver Section consists of 4 pole Analog Switch U8 (IC 4066) that demultiplexes the 4 Channels and the reconstruction unit.

The receiver timing logic is very similar to the transmitter timing logic. The demultiplexer based on the control signals C0, C1, C2, and C3 assigns the information to the corresponding channels. The success of the demultiplexer operation is fully dependent on how exactly RXCH0, RXCH1, RXCH2, RXCH3 signals match with the TXCH0, TXCH1, TXCH2, TXCH3 signals. Thus to ensure the proper demultiplexing, the two dividers are reset by the RXSYNC signals, which corresponds with the TXSYNC. (TXSYNC transmitter frame sync pulse and RXSYNC receiver frame sync pulse). The demultiplexed signals are then given to the corresponding reconstruction units.

The signal reconstruction unit is a 4th order Active Low Pass Butterworth Filter U16 (IC LM324) provided for each receiver channel. They filter out the sampling frequency and their harmonics from the demultiplexed signal and recover the base band by an integrate action. The cut-off frequency of the 4th Order Low Pass Butterworth Filter is 3.4 KHz.

A PLL U17 (IC4046) is used at the receiver for the recovery of the clock from the TXCH0. By this method, one line can be effectively conserved.

The TXSYNC has approximately 20% duty cycle and has a frequency equal to 1/4th of the clock. The clock is recovered from this signal by using PLL in the frequency multiplier mode. The multiplication factor is 4.

A standard digital phase locked loop CD4046 forms the heart of this logic and the Johnson counter CD4017 is used as a counter in the feedback path. RX clock for

the synchronization of samples and RXSYNC for the channel synchronization is recovered from the phase locked loop and Johnson counter respectively.

At the receiver a threshold detector is used to detect the DC level alone. This DC level is maximum in amplitude than other channel signal amplitude. DC level which is transmitted in the first channel (of the four available channels) occupies 20% of the frame, the threshold detector output is the same as the TXSYNC except that the duty cycle of the threshold detector output may be slightly greater than the TXSYNC.

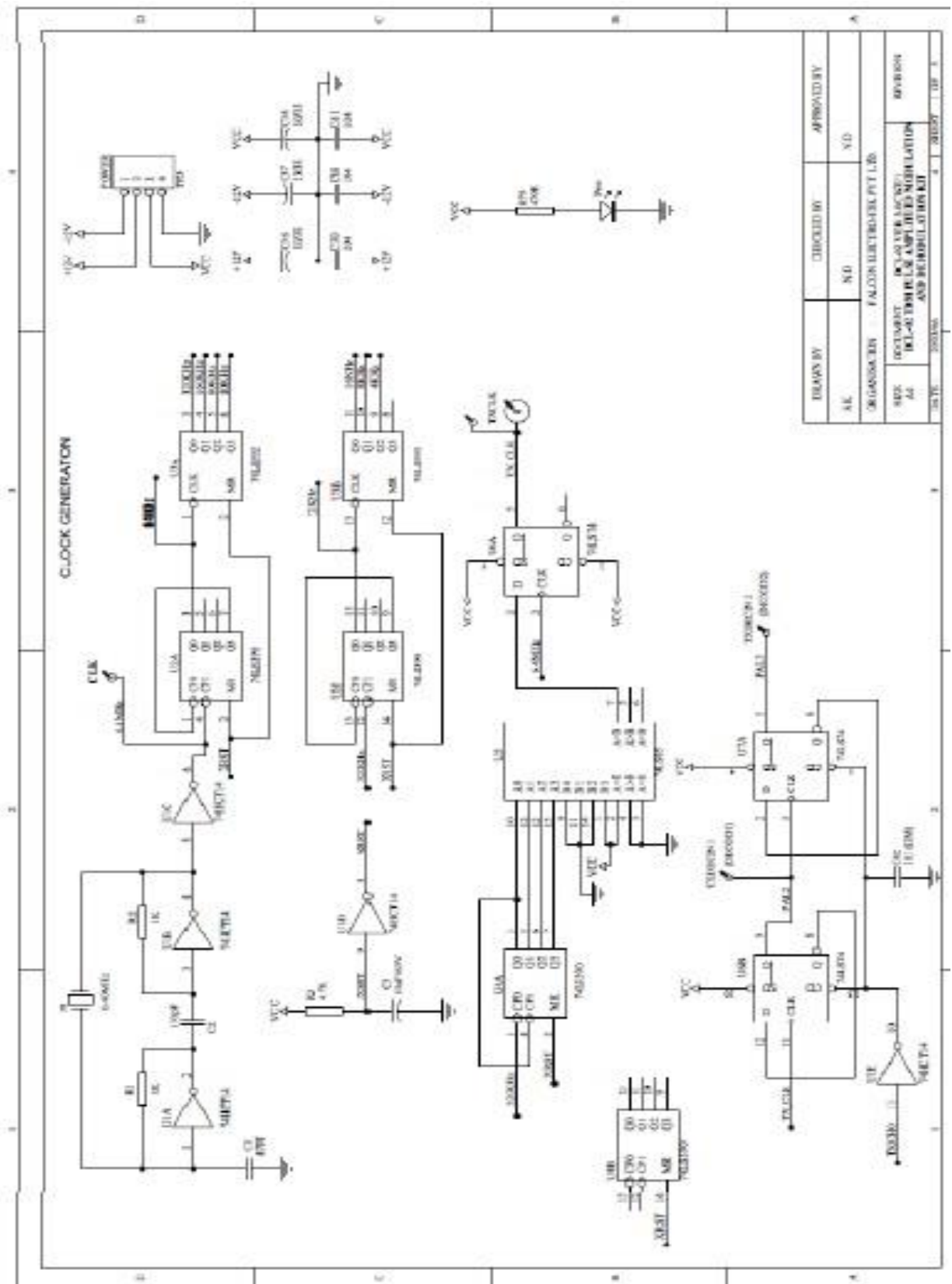
If this threshold detector output is used as the input to the above mentioned frequency multiplier logic built around CD4046BE and CD4017BE, then both the RXSYNC and clock can be recovered from the received data. The advantage of this scheme is that it utilizes a single wire communication with a common ground the disadvantage is that one separate channel has to be allocated for the DC level in this scheme.

## **SWITCH FAULTS**

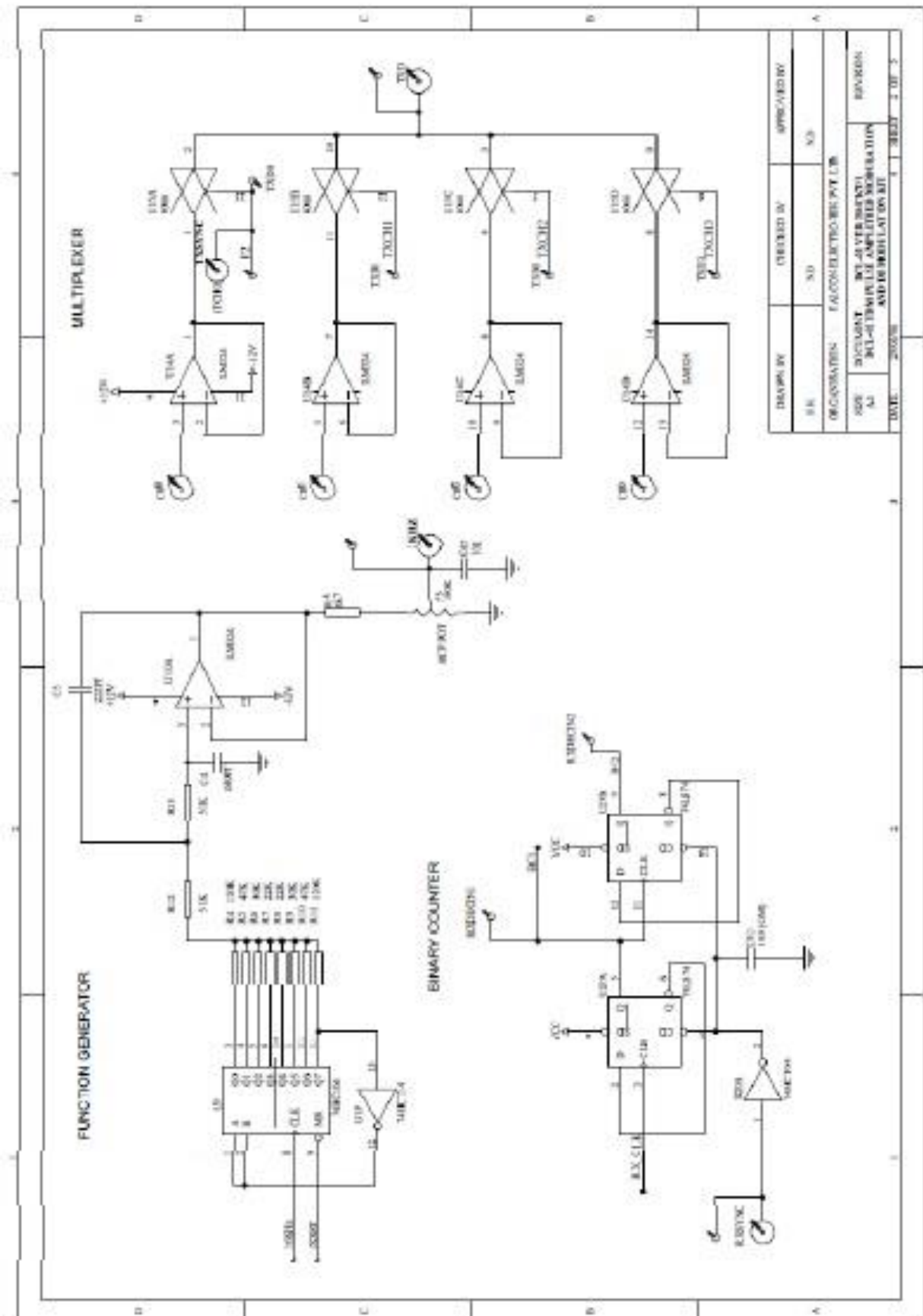
Switch fault section provides 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. The faults are normally used one at a time. To ensure that the DCL-02 kit is fully operational, all switches should be set to OFF position before use.

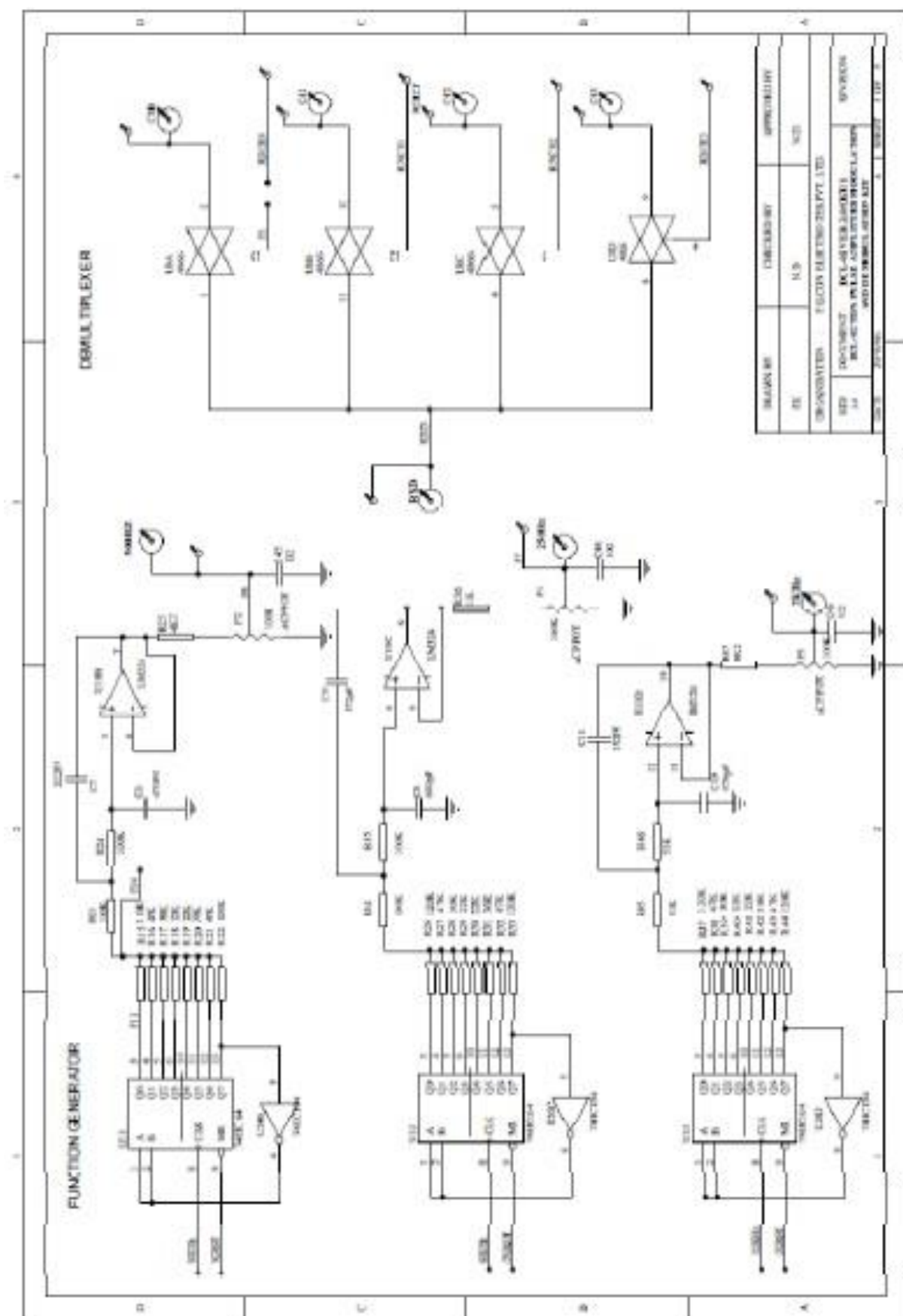
- Put switch **1** of **SF1** in Switch Fault section to **ON** position. This will short circuit 250Hz & 500Hz sine waves. We will get mixing of both the signals.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will change duty cycle for PLL recovery through Mono shot. This will mismatch the receiver demux pulses. Demultiplexed signals will not show synchronization.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will change reference voltage level of comparator. Hence output gets distorted only in level three connection of the circuit. The required and proper output can be achieved by setting threshold level.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short MSB of ladder network used for 500Hz sine wave generation. Shape of this sine wave changes.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will remove TXCH0 signal. This will remove all receiver-decoding pulses. Receiver outputs are disturbed.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will remove control signal of first channel in demultiplexer section. Output for Channel Zero is mixing of all signals.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will remove reset pulse for counter used to synchronize channel sequence. All channels get mixed.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will remove bypass capacitor from filter of third channel. Distorted output at channel three.

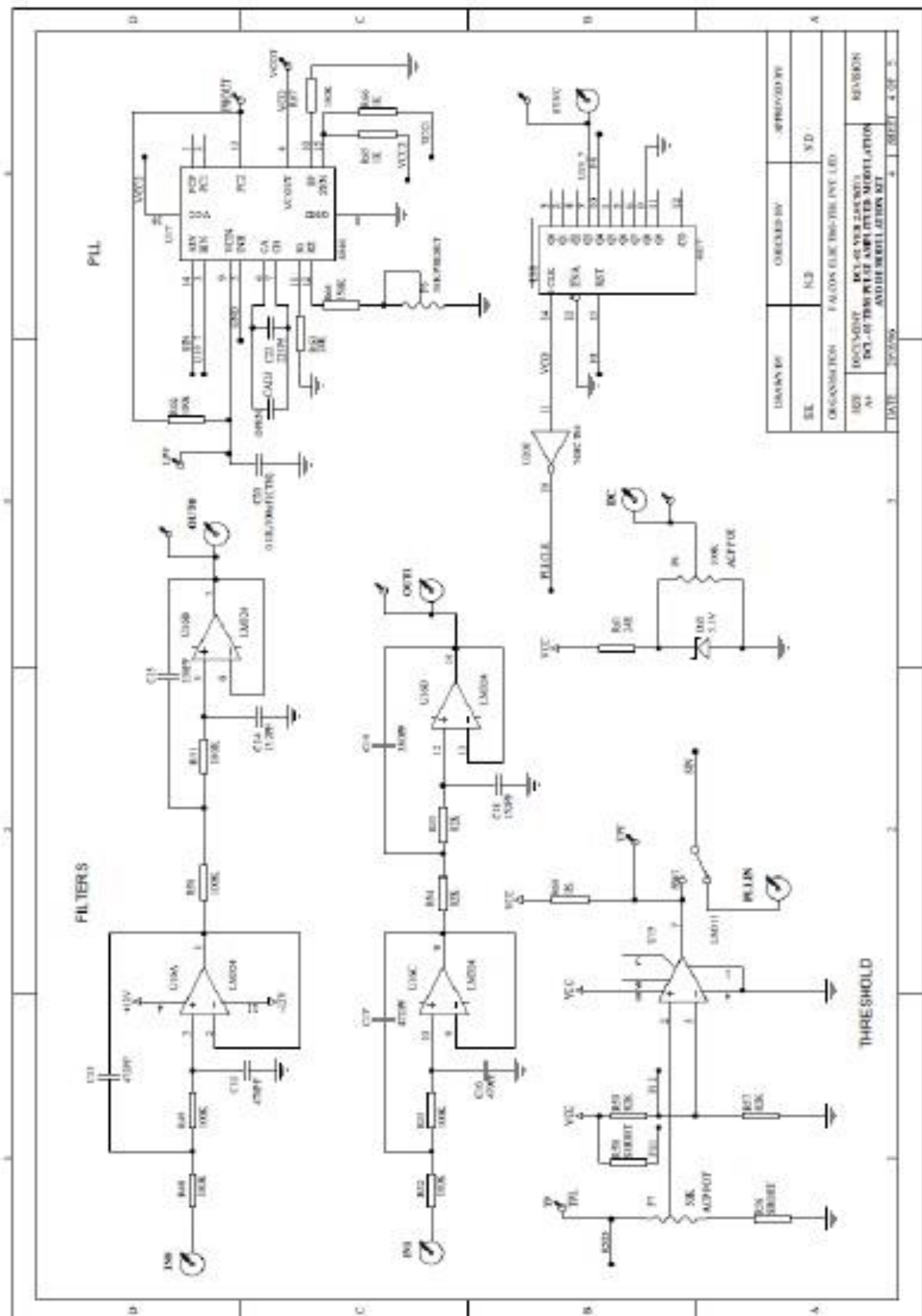
## CIRCUIT DIAGRAM

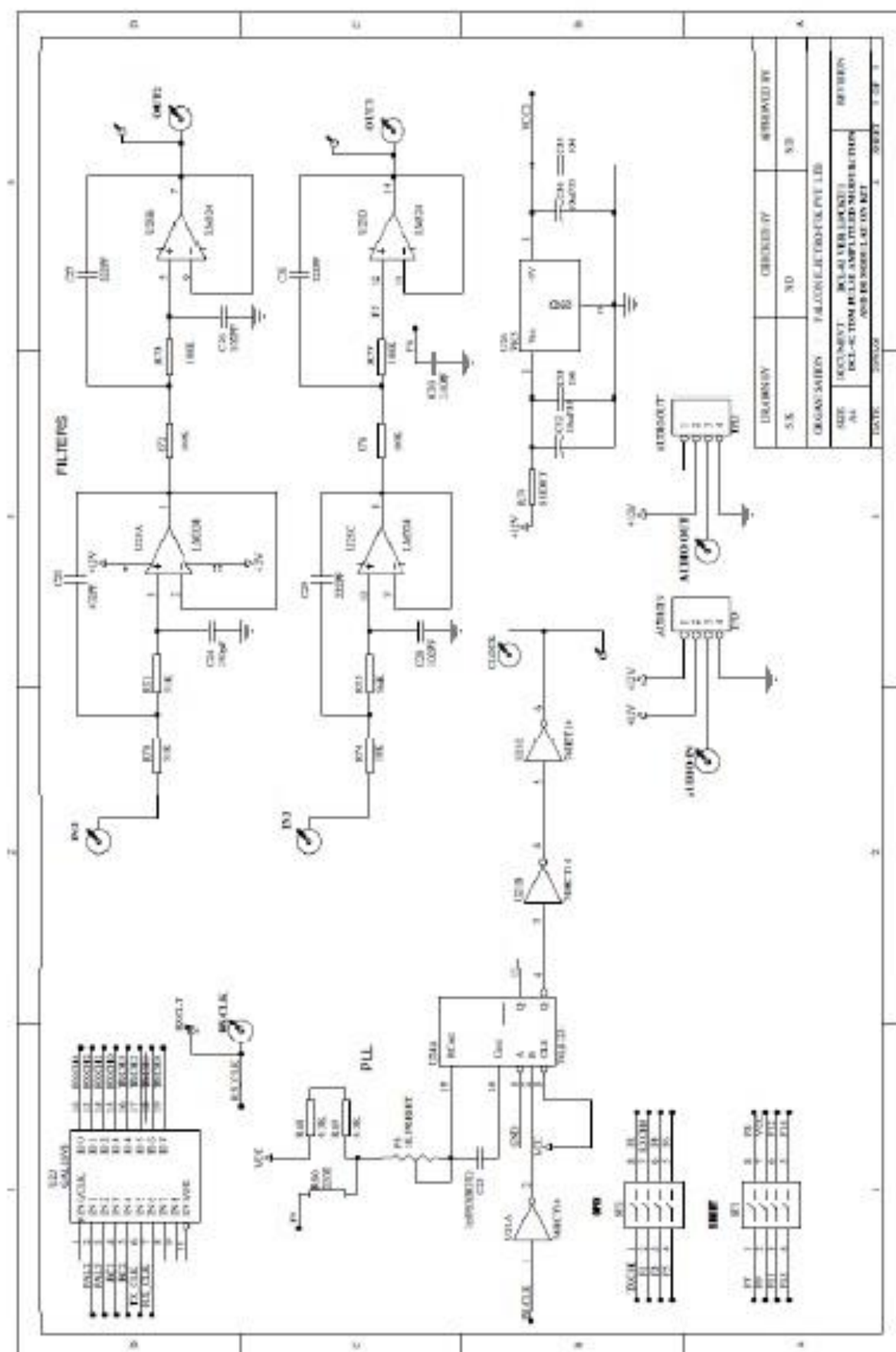




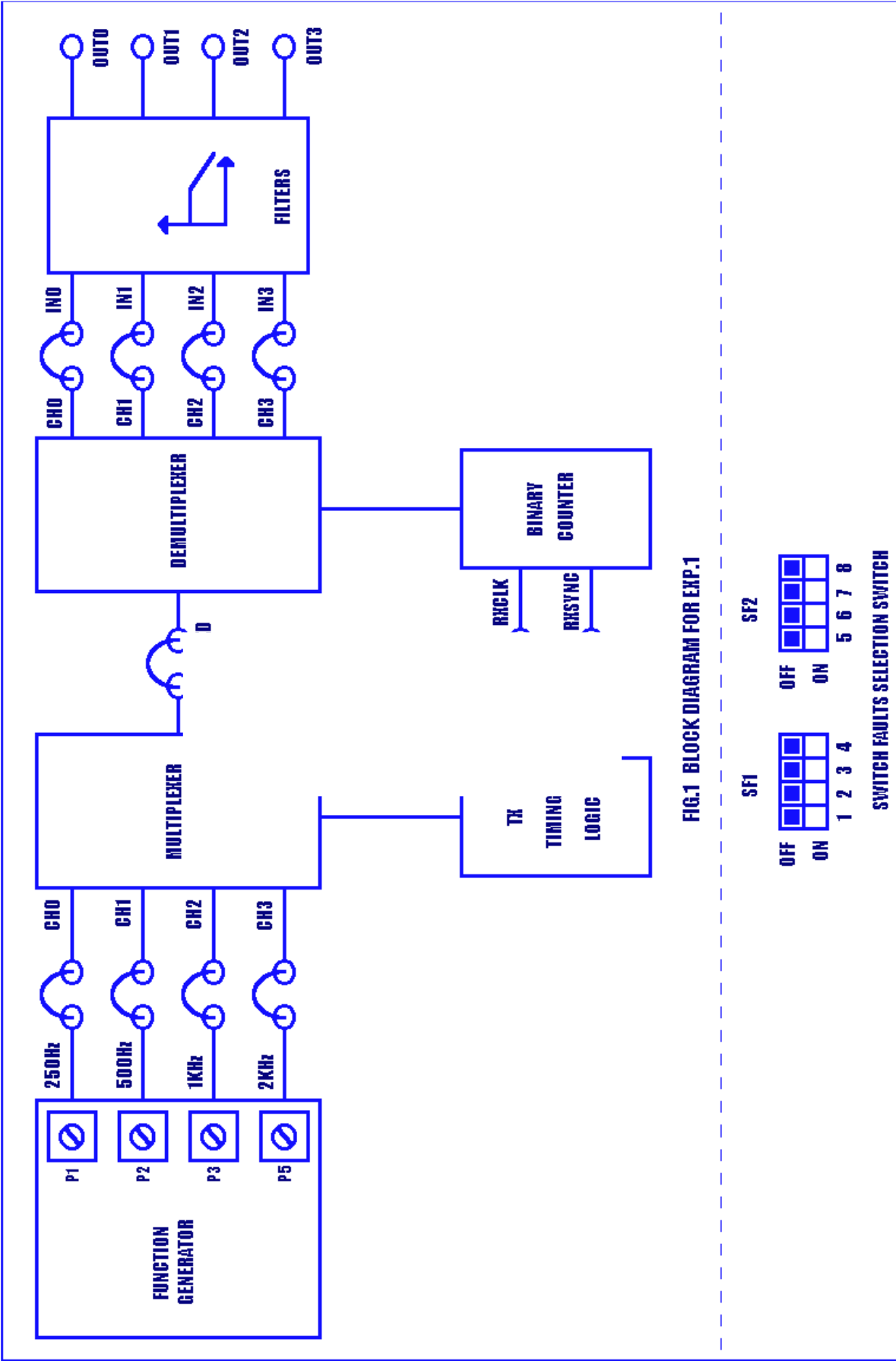








# **EXPERIMENT NO. 1**



## **EXPERIMENT NO: 1**

### **NAME:**

Study Of TDM Pulse Amplitude Modulation/Demodulation With Transmitter Clock And Channel Identification Information Linked Directly To The Receiver.

### **OBJECTIVE:**

To study Time Division Multiplexing and Demultiplexing, using Pulse Amplitude Modulation and Demodulation and to reconstruct the signals at the Receiver using Filters. The Transmitter Clock and the Channel Identification Information is linked directly to the Receiver.

### **THEORY:**

This module basically consists of the following sections:

- a. The Onboard Function Generator,
- b. The Transmitter,
- c. The Receiver with the associated synchronization circuitry.

#### **Onboard Function Generator**

This basically provides four Amplitude Variable each (0 - 5 V) synchronized sine waves, each 250Hz, 500Hz, 1KHz, and 2KHz and an amplitude variable DC level (0-5V). For the generation of sine waves please refer to circuit description manual.

#### **Transmitter**

The Transmitter Section consists of four Analog Input signals from the Function generator fed to the four channels of the Multiplexer where the signals fed are Time Division Multiplexed after undergoing the sampling. The sampling process makes the signals Pulse Amplitude Modulated. The frequencies for sampling are given from the decoder.

#### **Receiver**

The Receiver Section consists of a Demultiplexer that demultiplexes the four Time Division Multiplexed signals, which it receives from the transmitter. This Demultiplexed signals are then fed to the reconstruction circuit, which is the filter section.

The receiver timing logic is very similar to the transmitter timing logic. The demultiplexer based on the control signals C0, C1, C2, C3 assigns the information to the corresponding channels. The success of the de-multiplexer operation is fully dependent on how exactly, RXCH0, RXCH1, RXCH2, RXCH3 signals match with the TXCH0, TXCH1, TXCH2, TXCH3 signals. Thus, to ensure the proper de-multiplexing, two dividers are reset by the RXCH0 signal, which corresponds with the TXCH0. The demultiplexed signals are then given to the corresponding reconstruction units.

The signal reconstruction unit is a 4th order Active Low Pass Butterworth Filter provided for each receiver channel. They filter out the sampling frequency and their harmonics from the demultiplexed signal and recover the base band by an integrate action. The cut-off frequency of the 4th Order Low Pass Butterworth Filter is 3.4 KHz.

### **EQUIPMENTS:**

- DCL-02 Kit.
- Connecting Chords
- Power supply
- 20 MHz Dual Trace Oscilloscope

**NOTE:** Keep All The Switch Faults In Off Position.

## **PROCEDURE:**

1. Refer to the Block Diagram (Fig. 1) & Carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-02** & switch it on.
3. Connect **250Hz**, **500Hz**, **1 KHz**, and **2 KHz** sine wave signals from the Function Generator to the multiplexer inputs channel **CH0**, **CH1**, **CH2**, **CH3** by means of the connecting chords provided.
4. Connect the multiplexer output **TXD** of the transmitter section to the demultiplexer input **RXD** of the receiver section.
5. Connect the output of the receiver section **CH0**, **CH1**, **CH2**, **CH3** to the **IN0**, **IN1**, **IN2**, and **IN3** of the filter section.
6. Connect the sampling clock **TX CLK** and Channel Identification Clock **TXSYNC** of the transmitter section to the corresponding **RX CLK** and **RX SYNC** of the receiver section respectively.
7. Set the amplitude of the input sine wave as desired.
8. Take observations as mentioned below.

## **OBSERVATIONS:**

- Observe the following waveforms on oscilloscope and plot it on the paper.
- Input Channel **CH0**, **CH1**, **CH2**, **CH3**.
- Channel Selection Signal.
- **TX CLK** and **RX CLK**.
- Channel Identification Signal **TX SYNC** and **RX SYNC**.
- Multiplexer Output **TXD**.
- Demultiplexer Input **RXD**.
- Demultiplexer output **CH0**, **CH1**, **CH2**, **CH3**
- Reconstructed signal **OUT 0**, **OUT 1**, **OUT 2**, **OUT 3**

## **SWITCH FAULTS:**

**Note:** Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. These will short circuit 250Hz & 500Hz sine waves. We will get mixing of both the signals.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short MSB of ladder network used for 500Hz sine wave generation. Shape of this sine wave changes.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will remove TXCH0 signal. This will remove all receiver-decoding pulses. Receiver outputs are disturbed.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will remove control signal of first channel in demultiplexer section. Output for Channel Zero is mixing of all signals.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will remove bypass capacitor from filter of third channel. Distorted output at channel three.

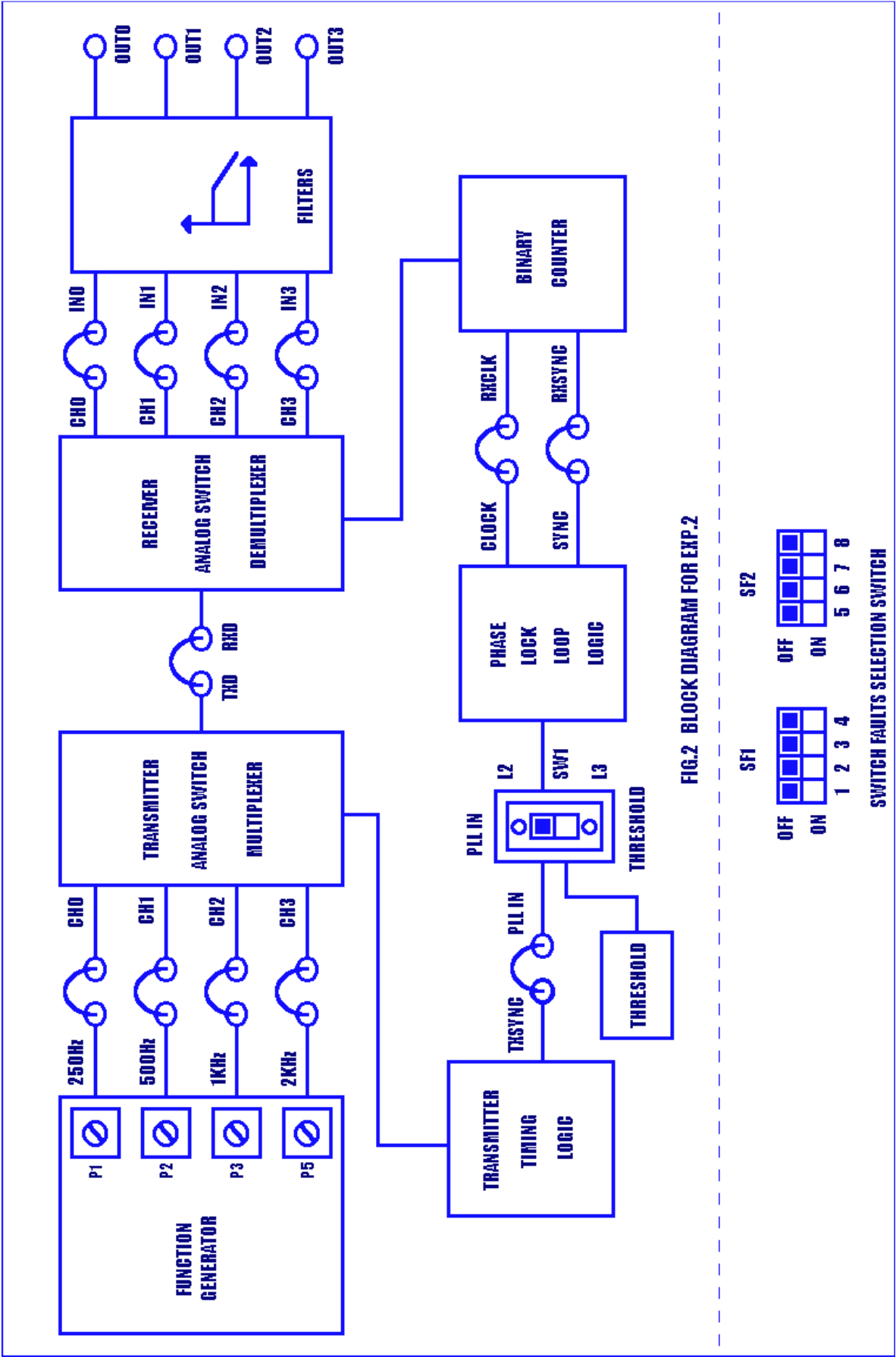


**CONCLUSION:**

In this experiment, the transmitter clock and the channel identification clock (Sync) are directly linked to the receiver section. Hence transmitter and receiver are synchronized and proper reconstruction of the signal is achieved.



# **EXPERIMENT NO. 2**



## **EXPERIMENT NO: 2**

### **NAME:**

Study Of TDM Pulse Amplitude Modulation / Demodulation With Channel Identification Information (Sync) Linked Directly To The Receiver.

### **OBJECTIVE:**

To study Time Division Multiplexing and Demultiplexing, using Pulse Amplitude Modulation and Demodulation and to reconstruct the signals at the Receiver using Filters. The Transmitter Channel Identification Information (sync) is only linked directly to the Receiver and the Receiver Clock and Channel Identification Information is now generated by the Phase Locked Loop circuitry.

### **THEORY:**

In TDM-PAM systems, for proper recovery of information, the TX clock and RX clock, TXSYNC and RXSYNC should be perfectly synchronized. For this purpose, the TX clock and TXSYNC can be transmitted along with the data but it calls for another two lines of transmission. For conserving the line, a PLL is used at the receiver for the recovery of the clock from the TXSYNC. By this method, one line can be effectively conserved.

### **EQUIPMENTS:**

- DCL-02 Kit.
- Connecting Chords
- Power supply
- Dual Trace Oscilloscope

**NOTE:** Keep All The Switch Faults in Off Position.

### **PROCEDURE:**

1. Refer to the Block Diagram (Fig. 2) & Carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-02** & switch it on.
3. Connect **250Hz**, **500Hz**, **1 KHz**, and **2 KHz** sine wave signals to the multiplexer input channel **CH0**, **CH1**, **CH2** and **CH3** by means of the Connecting chords provided.
4. Connect the multiplexer output **TXD** of the transmitter section to the demultiplexer input **RXD** of the receiver section.
5. Connect the output of the Demultiplexer **CH0**, **CH1**, **CH2** and **CH3** to the input of the filter section **IN 0**, **IN 1**, **IN 2**, **IN 3**.
6. Connect the Channel Identification Signal **TXSYNC** of the transmitter section to the PLL Input post **PLL IN**.
7. Set the PLL Input Switch **SW1** to **L2** position.
8. Connect the **SYNC** and **CLOCK** generated by the PLL circuitry to **RXSYNC** and **RXCLK** of the receiver respectively.
9. Set the amplitude of the input sine wave as desired.

10. Take observations as mentioned below.
11. If the reconstructed signal appears to be improper press **RESET** switch to get proper signal reconstruction.

### **OBSERVATIONS:**

- Observe the following waveforms on oscilloscope and plot it on the paper.
- Input Channel **CH0, CH1, CH2, CH3**
- Transmitter Clock **TX CLK**.
- Channel Identification Signal **TX SYNC**.
- Multiplexer Output **TXD**.
- Demultiplexer Input **RXD**.
- SYNC and CLOCK generated by **PLL**.
- Demultiplexer output **CH0, CH1, CH2 and CH3**.
- Reconstructed signal **OUT 0, OUT 1, OUT 2, OUT 3**.

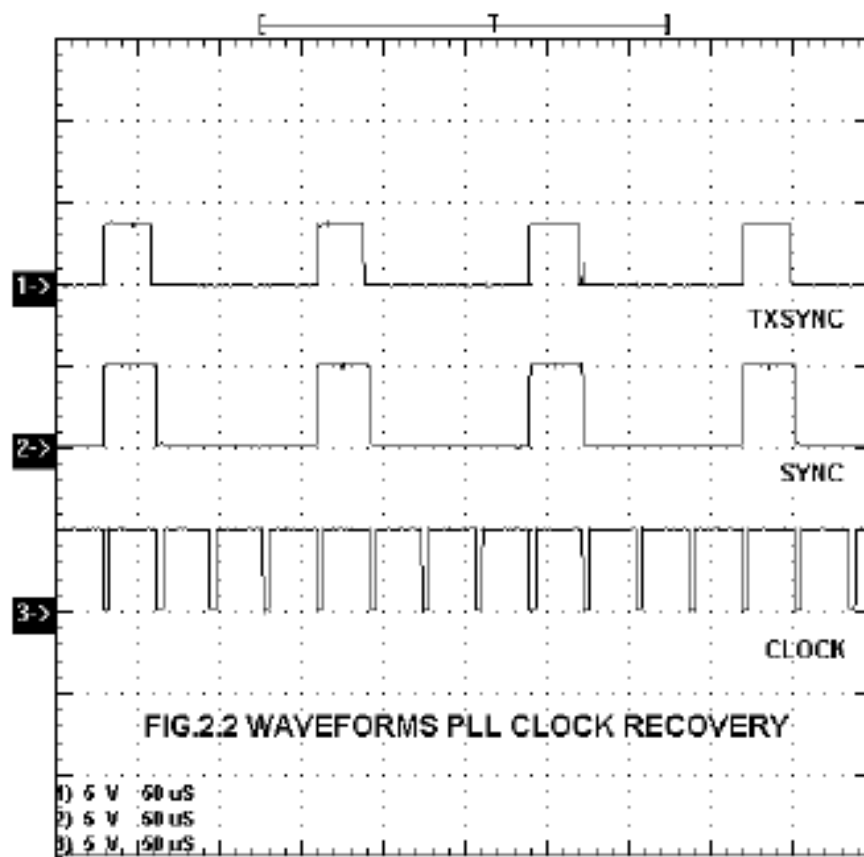
### **SWITCH FAULTS:**

**Note:** Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. These will short circuit 250Hz & 500Hz sine waves. We will get mixing of both the signals.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will change duty cycle for PLL recovery through Monoshot. This will mismatch the receiver demux pulses. Demultiplexed signals will not show synchronization.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short MSB of ladder network used for 500Hz sine wave generation. Shape of this sine wave changes.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will remove TXCH0 signal. This will remove all receiver-decoding pulses. Receiver outputs are disturbed.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will remove control signal of first channel in demultiplexer section. Output for Channel Zero is mixing of all signals.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will remove reset pulse for counter used to synchronize channel sequence. All channels get mixed.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will remove bypass capacitor from filter of third channel. Distorted output at channel three.

### **CONCLUSION:**

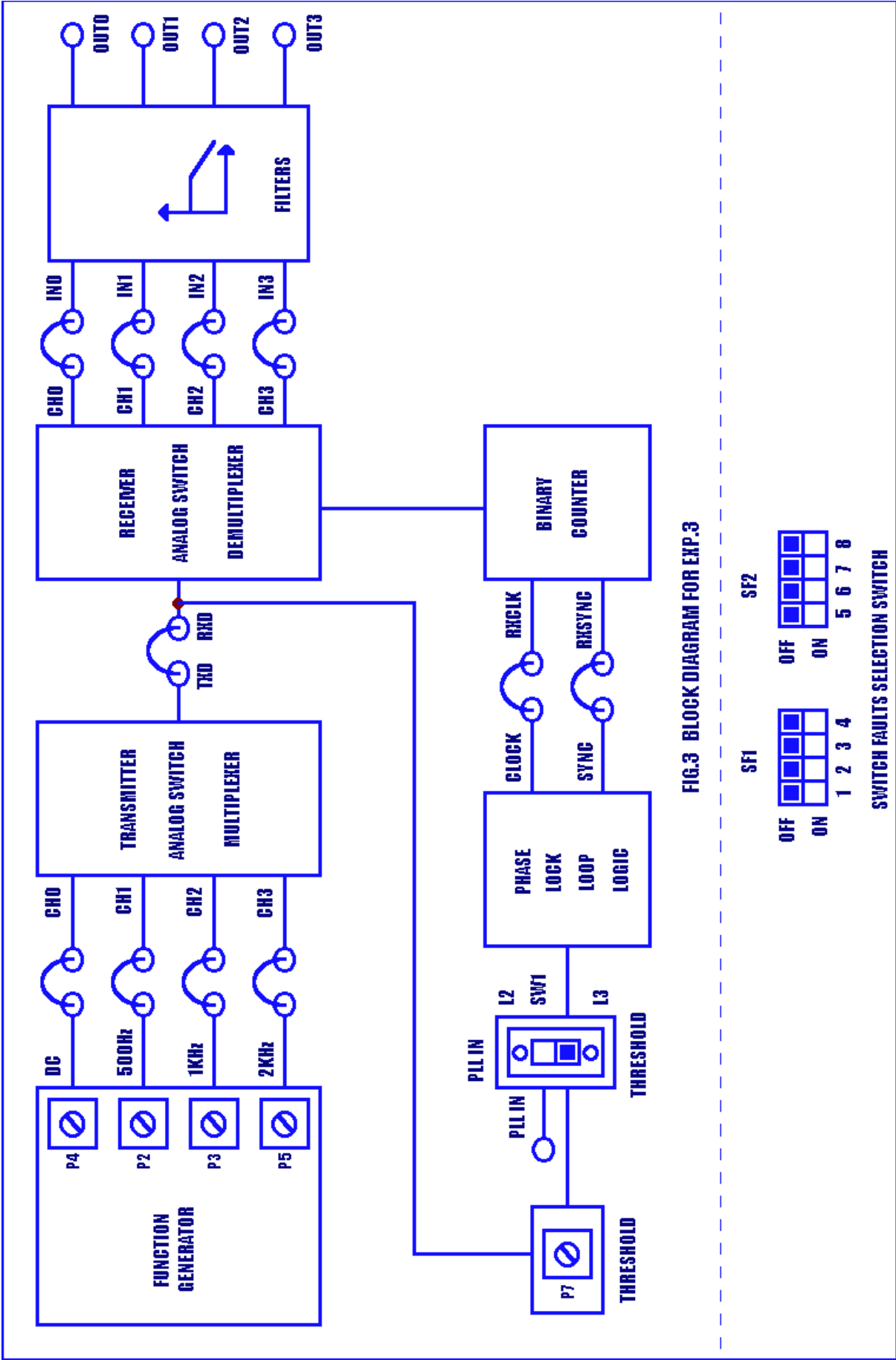
In this experiment, the channel identification signal is linked to the PLL input of the receiver section. Here the PLL locks on the TXSYNC Signal Frequency and the PLL along with its associated circuitry generates the SYNC and CLOCK Signals, which are same in Phase and Frequency as that of the Transmitter TXSYNC and TXCLK Signal and hence synchronize the Transmitter to the Receiver and proper reconstruction of the signal is achieved.







# **EXPERIMENT NO. 3**



## **EXPERIMENT NO: 3**

### **NAME:**

To Study The Use Of A Threshold Level Comparator And Phase Locked Loop Circuitry To Generate The Receiver Clock And Channel Identification Information.

### **OBJECTIVE:**

To study Time Division Multiplexing and Demultiplexing, using Pulse Amplitude Modulation and Demodulation and to reconstruct the signals at the Receiver using Filters. The Receiver derives the Clock and Channel Identification Information from the incoming stream of Time Division Multiplexed Pulses. The first channel of Transmitter is used to send the synchronization pulse. Using the Threshold Level Comparator and Phase-locked Loop Circuitry, the Receiver Clock and Channel Identification Information is obtained.

### **THEORY:**

The RX CLK and RXSYNC can be recovered from the transmitted data itself, if we transmit a DC level as information in the first channel. The DC level should have amplitude which is greater than the amplitude of the information signals that are transmitted in other channels. At the receiver a threshold detector is used to detect the DC level alone. The advantage of this scheme is that it utilizes a single wire communication with a common ground, the disadvantage is that one separate channel had to be allocated for the DC level in this scheme.

### **EQUIPMENTS:**

- DCL-02 Kit.
- Connecting Chords
- Power supply
- 20 MHz Dual Trace Oscilloscope

**NOTE:** Keep All The Switch Faults In Off Position.

### **PROCEDURE:**

1. Refer to the Block Diagram (Fig. 3) & Carry out the following connections and switch settings. Connect power supply in proper polarity to the kit **DCL-02** & switch it on.
2. Connect **DC** signal to the first channel of the multiplexer **CH0**.
3. Connect the **500Hz**, **1 KHz**, and **2 KHz** sine wave signals to the multiplexer input channel **CH1**, **CH2**, **CH3**.
4. Keep the amplitude of **DC** signal at maximum using **P4** completely clockwise.
5. Keep the amplitude of other signals around **2 Volts** not more than that.
6. Connect the multiplexer output **TXD** of the transmitter section to the demultiplexer input **RXD** of the receiver section.
7. Connect the output of the Receiver **CH0**, **CH1**, **CH2** and **CH3** to the input of the filter section **IN 0**, **IN 1**, **IN 2**, **IN 3**.
8. Set the PLL Input Switch **SW1** to **L3** position.
9. Connect the **SYNC** and **CLOCK** generated by the PLL circuitry to **RXSYNC** and **RXCLK** of the receiver respectively.
10. Set the amplitude of the input sine wave lower than the amplitude of the **SYNC LEVEL**.

11. Vary the Threshold Level Comparator potentiometer **P7**, so that the DC level transmitted, is recovered without distortion in the first channel of the receiver.
12. Take observations as mentioned below.

## **OBSERVATIONS**

Observe the following waveforms on oscilloscope and plot it on the paper.

- Input Channel **CH0, CH1, CH2, CH3**.
- Transmitter Clock **TX CLK**.
- Channel Identification Signal **TX SYNC**.
- Multiplexer Output **TXD**.
- Demultiplexer Input **RXD**.
- **SYNC** and **CLOCK** generated by PLL.
- Demultiplexer output **CH0, CH1, CH2, CH3**
- Reconstructed signal **OUT 0,OUT 1,OUT 2,OUT 3**

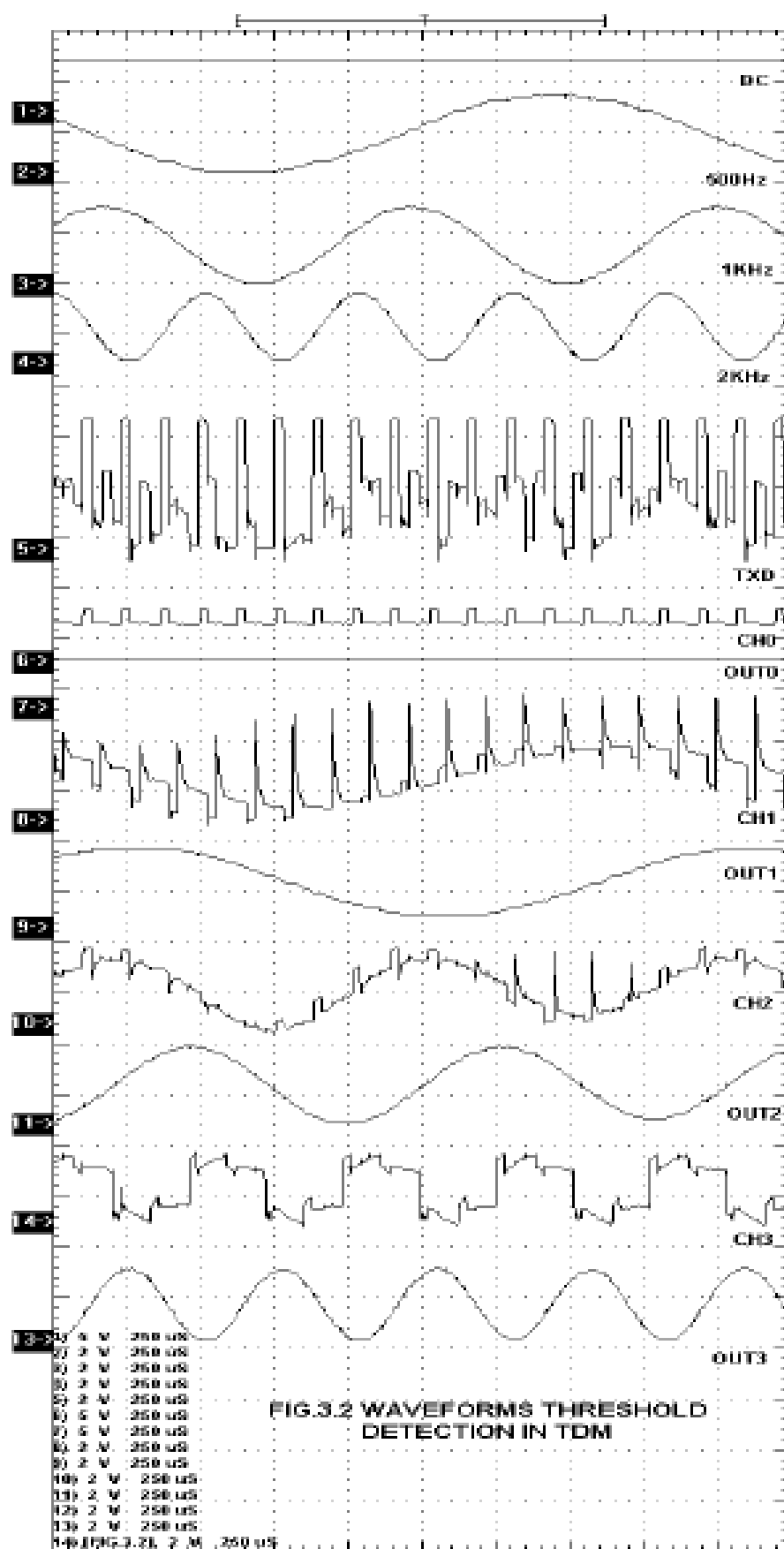
## **SWITCH FAULTS**

**Note:** Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

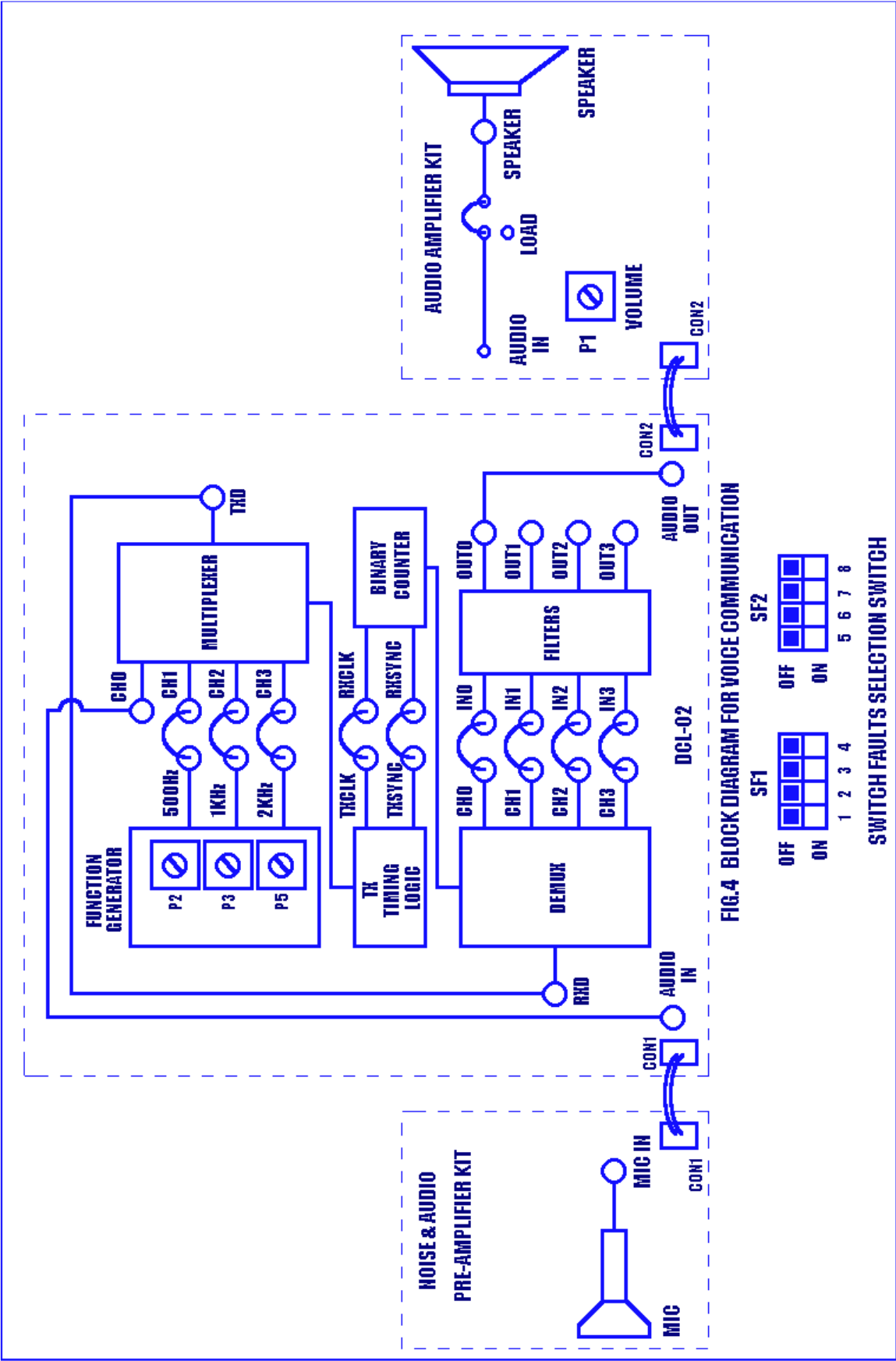
- Put switch **1** of **SF1** in Switch Fault section to **ON** position. These will short circuit 250Hz & 500Hz sine waves. We will get mixing of both the signals.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will change duty cycle for PLL recovery through Mono shot. This will mismatch the receiver demux pulses. Demultiplexed signals will not show synchronization.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will change reference voltage level of comparator. Hence output gets distorted only in level three connection of the circuit. The required and proper output can be achieved by setting threshold level.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short MSB of ladder network used for 500Hz sine wave generation. Shape of this sine wave changes.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will remove TXCH0 signal. This will remove all receiver-decoding pulses. Receiver outputs are disturbed.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will remove control signal of first channel in demultiplexer section. Output for Channel Zero is mixing of all signals.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will remove reset pulse for counter used to synchronize channel sequence. All channels get mixed.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will remove bypass capacitor from filter of third channel. Distorted output at channel three.

## **CONCLUSION**

The PLL will lock into the frequency of generation of the sync pulses and the PLL along with the associated circuitry will generate the sync and clock signals, which are same in phase and frequency as that of the transmitter TXSYNC, and TXCLOCK signal, hence synchronizing the transmitter to the receiver.



# **EXPERIMENT NO. 4**





## **EXPERIMENT NO: 4**

### **NAME:**

Voice Communication

### **OBJECTIVE:**

Study of voice communication using Pulse Amplitude Modulation with channel identification information using the Threshold circuit and PLL

### **THEORY:**

Voice being the Analog signal can be modulated using different methods, among this one method is the Pulse Amplitude Modulation. In this we study the time division multiplexing of the modulated signal. For performing the Voice communication, set up the basic Experiment mentioned earlier. Voice signal is transmitted by interleaving samples of other channel signals. At the receiver section it is demultiplexed from other channel signals. After processing through Audio Amplifier we can get audio output.

### **EQUIPMENTS:**

- Experiment kit DCL-02.
- Connecting Chords
- Power supply
- 20 MHz Dual Trace Oscilloscope
- Microphone
- Noise and Audio Pre-Amplifier kit
- Audio Amplifier kit
- Power connection cables

**NOTE:** Keep All The Switch Faults In Off Position.

### **PROCEDURE:**

1. Refer to the Block Diagram (Fig. 4) & Carry out the following connections and switch settings.
2. Connect the Noise and Audio Pre-Amplifier kit to the **CON1** Post on **DCL-02** kit using power connection cable.
3. Connect the Audio Amplifier kit to the **CON2** Post on **DCL-02** kit using power connection cable.
4. Connect Microphone to the Noise and Audio Pre-Amplifier kit at **MIC IN** post.
5. Connect the input signal from **AUDIO IN** Post to **CH 0** of the multiplexer.
6. Connect the **500Hz, 1 KHz, and 2 KHz** sine wave signal to the multiplexer input channel **CH1, CH2, CH3**.
7. Connect the sampling clock **TX CLK** and Channel Identification Clock **TXSYNC** of the transmitter section to the corresponding **RX CLK** and **RX SYNC** of the receiver section respectively.
8. Connect the output of the Receiver **CH0, CH1, CH2 and CH3** to the input of the filter section **IN 0, IN 1, IN 2 and IN 3**.

9. Check the voice signal at the out points of the filter. At **OUT 0** posts you will find the voice signal connect this signal **OUT 0** post to the **AUDIO OUT**.
10. Connect the speaker to the speaker jack on Audio Amplifier Kit.
11. Repeat voice link experiment using the procedure for level two and level three experiments.

## **SWITCH FAULTS:**

**Note:** Keep the connections as per the procedure. Now switch corresponding fault switch button in ON condition & observe the different effect on the output. The faults are normally used one at a time.

- Put switch **1** of **SF1** in Switch Fault section to **ON** position. These will short circuit 250Hz & 500Hz sine waves. We will get mixing of both the signals.
- Put switch **2** of **SF1** in Switch Fault section to **ON** position. This will change duty cycle for PLL recovery through Mono shot. This will mismatch the receiver demux pulses. Demultiplexed signals will not show synchronization.
- Put switch **3** of **SF1** in Switch Fault section to **ON** position. This will change reference voltage level of comparator. Hence output gets distorted only in level three connection of the circuit. The required and proper output can be achieved by setting threshold level.
- Put switch **4** of **SF1** in Switch Fault section to **ON** position. This will short MSB of ladder network used for 500Hz sine wave generation. Shape of this sine wave changes.
- Put switch **5** of **SF2** in Switch Fault section to **ON** position. This will remove TXCH0 signal. This will remove all receiver-decoding pulses. Receiver outputs are disturbed.
- Put switch **6** of **SF2** in Switch Fault section to **ON** position. This will remove control signal of first channel in demultiplexer section. Output for Channel Zero is mixing of all signals.
- Put switch **7** of **SF2** in Switch Fault section to **ON** position. This will remove reset pulse for counter used to synchronize channel sequence. All channels get mixed.
- Put switch **8** of **SF2** in Switch Fault section to **ON** position. This will remove bypass capacitor from filter of third channel. Distorted output at channel three.